Long Division

Divide, multiply, take away, bring down. Repeat.

7,488 ÷ 24

74 ÷ 24 ≈ 3	3 × 24 = 72	74 − 72 = 2, bring down 8: 28
28 ÷ 24 ≈ 1	1 × 24 = 24	28 − 24 = 4, bring down 8: 48
48 ÷ 24 ≈ 2	2 × 24 = 48	48 − 48 = 0

7,488 ÷ 24 = 312 Each step is one digit of the answer.

Use easy parts to guess each digit (packet 78)

How many 24s in 74? 24 × 3 = 72 fits, 24 × 4 = 96 is too much. So 3.
If what is left at the end is not 0, write it as R: 500 ÷ 24 = 20 R 20.

Your turn!

936 ÷ 12 = 78

For teachers

Goal: divide multi-digit numbers with the standard algorithm (6.NS.2). Say “Divide, multiply, take away, bring down.” Kids can still use packet 78's easy parts to find each digit.

Long Division: Practice

Divide, multiply, take away, bring down.

Part 1 One-digit divisors Work under the number.

1) 213 / 4) 852

2) 226 / 6) 1,356

3) 549 / 5) 2,745

4) 306 / 3) 918

5) 512 / 8) 4,096

6) 506 / 7) 3,542

7) 208 / 9) 1,872

8) 1,243 / 5) 6,215

9) 1,167 / 2) 2,334

Part 2 Two-digit divisors Use easy parts to find each digit.

10) 312 / 24) 7,488

11) 168 / 31) 5,208

12) 399 / 23) 9,177

Long Division: Practice (continued)

Part 2 Two-digit divisors (continued)

13) 384) 18 6,912

14) 198) 42 8,316

15) 187) 25 4,675

16) 243) 14 3,402

17) 210) 47 9,870

18) 95) 63 5,985

Part 3 Divide No help this time.

19) 3,920 ÷ 16 = 245

20) 3,744 ÷ 32 = 117

21) 8,316 ÷ 27 = 308

22) 7,290 ÷ 45 = 162

23) 6,838 ÷ 13 = 526

24) 8,178 ÷ 58 = 141

Long Division: Practice (continued)

Part 3 Divide (continued)

25) 9,093 ÷ 21 = 433

26) 7,452 ÷ 36 = 207

27) 7,296 ÷ 19 = 384

28) 8,000 ÷ 64 = 125

29) 8,848 ÷ 28 = 316

30) 8,823 ÷ 51 = 173

Part 4 What is left? No help this time.

31) 2,000 ÷ 37 = 54 R 2

32) 5,431 ÷ 16 = 339 R 7

33) 7,777 ÷ 25 = 311 R 2

34) 3,210 ÷ 43 = 74 R 28

35) 9,001 ÷ 12 = 750 R 1

36) 6,543 ÷ 58 = 112 R 47

Long Division: Practice (continued)

Part 4 What is left? (continued)

37) $4{,}444 \div 33 = 134$ R 22

38) $8{,}123 \div 21 = 386$ R 17

Part 5 More long division

39) $13\overline{)1{,}326}$ = 102

40) $56\overline{)12{,}320}$ = 220

41) $30\overline{)5{,}490}$ = 183

42) $58\overline{)8{,}874}$ = 153

43) $14\overline{)2{,}044}$ = 146

44) $35\overline{)10{,}360}$ = 296

45) $13\overline{)598}$ = 46

46) $27\overline{)6{,}183}$ = 229

47) $48\overline{)12{,}288}$ = 256

Long Division: Practice (continued)

Part 6 More dividing

48) 6,260 ÷ 20 = 313

49) 15,120 ÷ 45 = 336

50) 8,004 ÷ 69 = 116

51) 17,520 ÷ 73 = 240

52) 1,386 ÷ 33 = 42

53) 3,904 ÷ 16 = 244

54) 8,225 ÷ 25 = 329

55) 1,620 ÷ 18 = 90

56) 4,704 ÷ 14 = 336

57) 1,820 ÷ 13 = 140

58) 7,062 ÷ 33 = 214

59) 8,671 ÷ 29 = 299

Long Division: Practice (continued)

Part 6 More dividing (continued)

60) $16{,}779 \div 51 = 329$

61) $5{,}720 \div 40 = 143$

62) $10{,}972 \div 52 = 211$

63) $23{,}247 \div 63 = 369$

64) $12{,}825 \div 45 = 285$

65) $21{,}560 \div 55 = 392$

Part 7 Remember? Problems from earlier packets.

66) $\frac{3}{4} \div \frac{1}{8} = 6$

67) 35% of 60 = 21

68) 72 pages in 4 days = 18 pages per day

69) $13 : 6 = 39 : 18$

70) $3.2 \times 5 = 16$

71) $\frac{7}{10} - \frac{1}{4} = \frac{9}{20}$

Long Division: Practice (continued)

Part 8 Word problems Divide, multiply, take away, bring down.

72) A school has 1,248 books on 16 shelves. Each shelf holds the same. How many books on a shelf?

Number sentence 1,248 ÷ 16 = 78

Answer 78 books

Draw or show your work

73) A farm packs 3,456 eggs in cartons of 12. How many cartons?

Number sentence 3,456 ÷ 12 = 288

Answer 288 cartons

Draw or show your work

74) A trip of 2,310 miles takes 42 hours. How many miles an hour is that?

Number sentence 2,310 ÷ 42 = 55

Answer 55 miles

Draw or show your work

75) 1,000 stickers are shared by 24 kids. How many does each kid get?

Number sentence 1,000 ÷ 24 = 41 R 16

Answer 41 stickers

Draw or show your work

76) Challenge: 1,000 people go to a game. A bus holds 45. How many buses are needed?

Number sentence 1,000 ÷ 45 = 22 R 10, so one more

Answer 23 buses

Draw or show your work

Check-In: Long Division

Show what you know. No help this time.

Mastery: 7 of 8 (80%)

Part 1 Long division

1) $26 \overline{)6{,}552}$ = 252

2) $8 \overline{)2{,}592}$ = 324

Part 2 Divide

3) $5{,}542 \div 17 =$ 326

4) $2{,}982 \div 14 =$ 213

5) $4{,}469 \div 41 =$ 109

6) $6{,}020 \div 35 =$ 172

Part 3 What is left?

7) $7{,}250 \div 34 =$ 213 R 8

8) $3{,}000 \div 29 =$ 103 R 13